

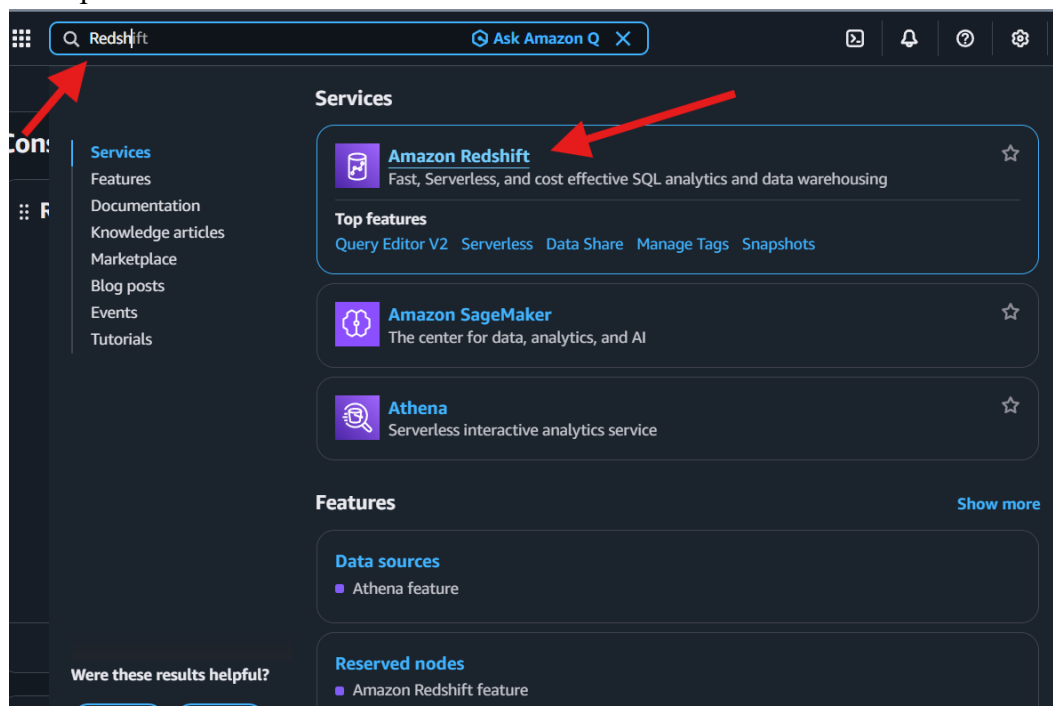
Querying an Amazon Redshift Cluster

To Begin with the Lab

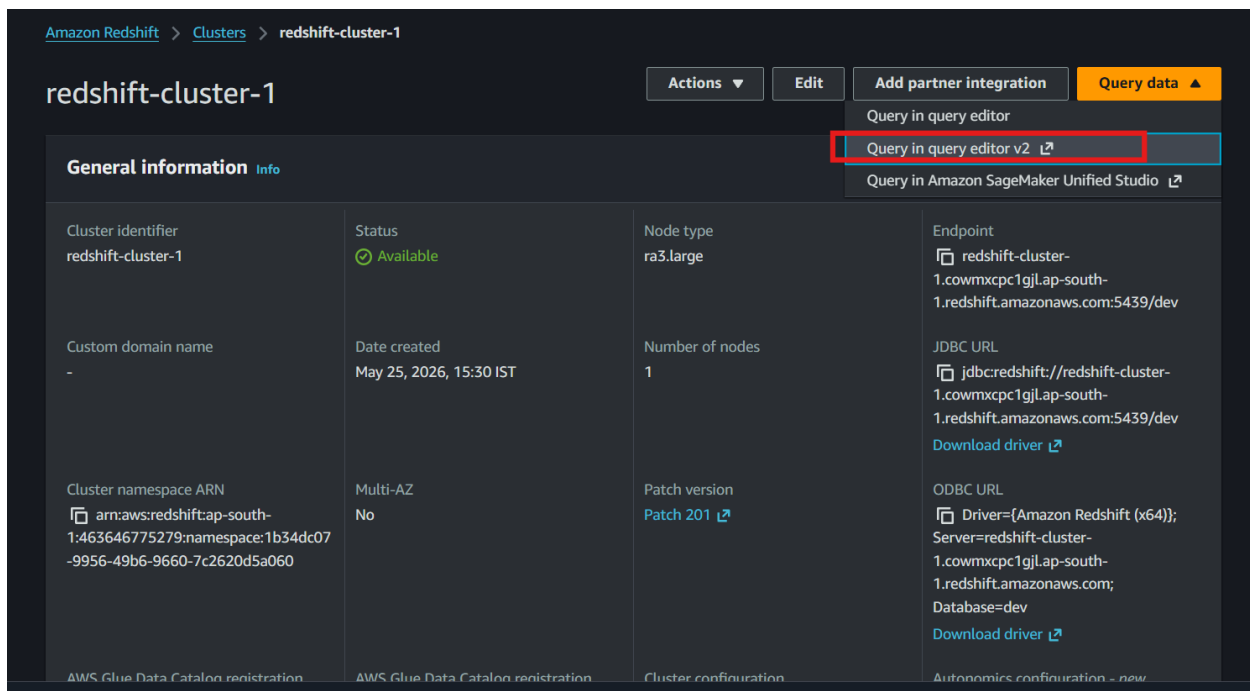
Summary of the Lab

In this lab, **Amazon Redshift Query Editor v2** was connected to an existing **Amazon Redshift** provisioned cluster using temporary database credentials. After establishing the connection, the available native and external databases were explored, and the Tickit sample dataset was reviewed within the public schema. A sample notebook was duplicated and used to execute SQL queries against the cluster. During query execution, a schema-related error was encountered because the queries referenced a non-existent Tickit schema. The issue was resolved by using the tables from the public schema, and successful query execution verified that the cluster connection and sample dataset were working correctly.

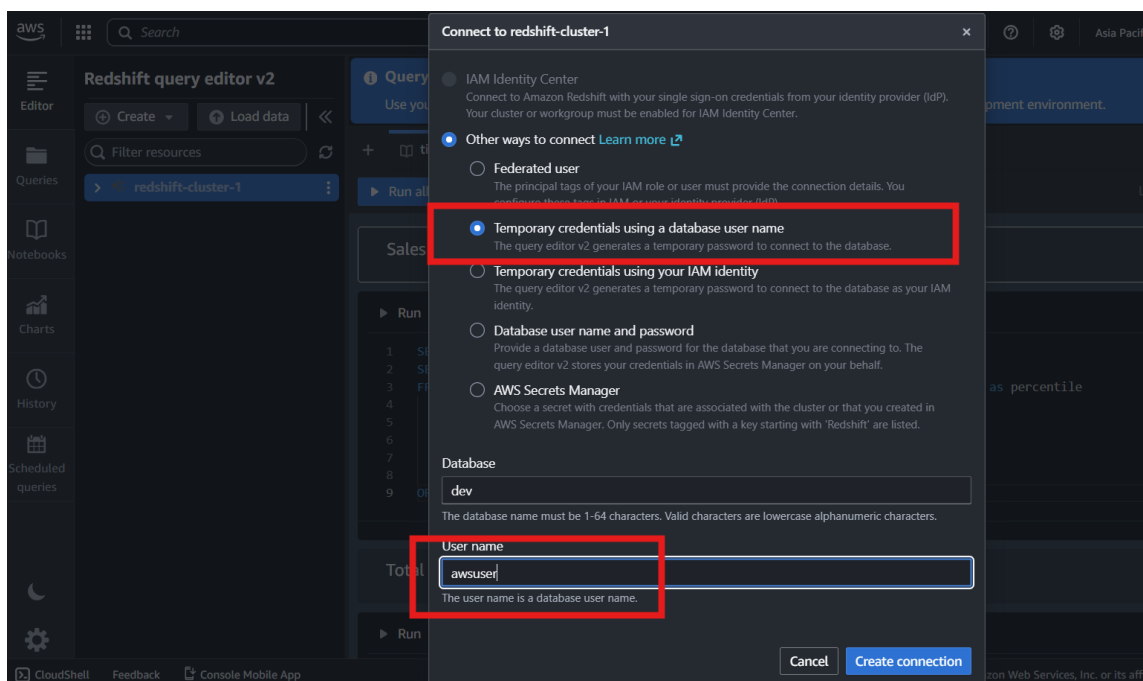
- Sign in to the AWS Management Console
- Search for **Amazon Redshift**
- Open **Amazon Redshift**



- Navigate to **Serverless dashboard**
- Open the previously created cluster
- Click **Query data** → **Query in query editor v2**
- It will open **Query Editor v2**

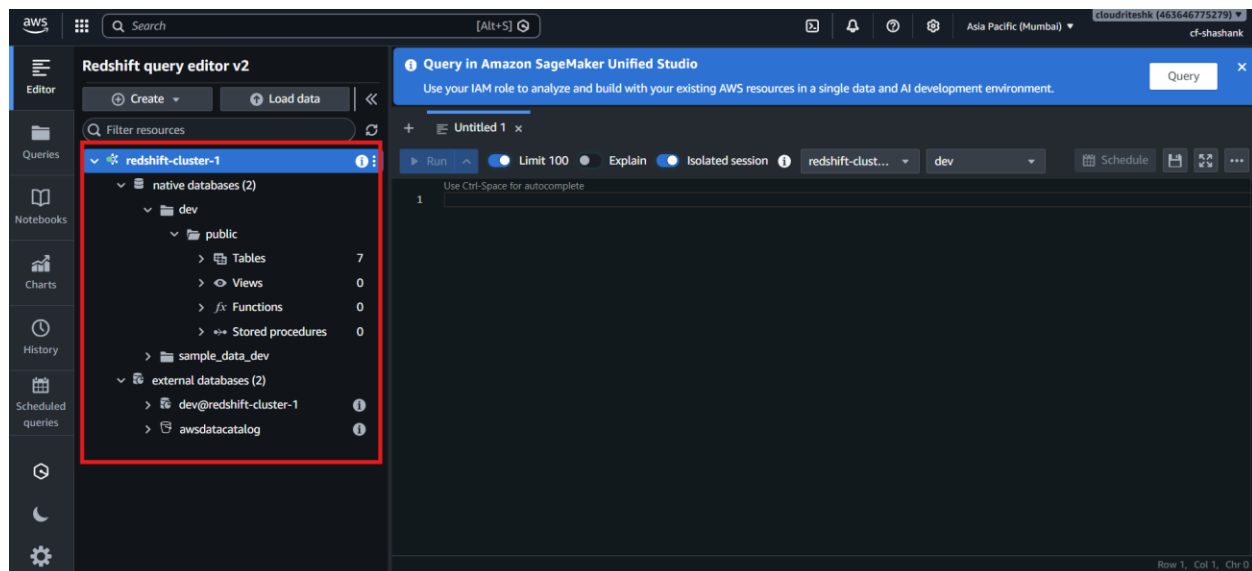


- Select **Temporary credentials using database username**
- Enter the database name as **dev**
- Enter the username as **awsuser**
- Click **Create connection**

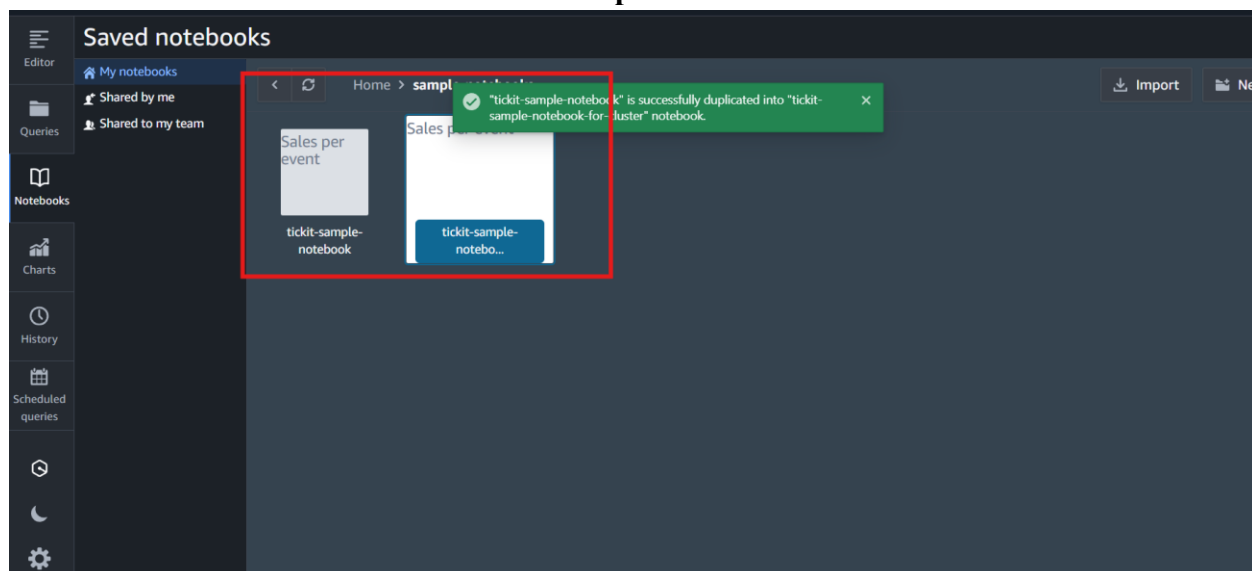


- Verify that the cluster connection is established successfully
- Review the available native databases
- Verify that **dev** and **sample_data_dev** databases are available
- Review the available external databases

- Verify that **awsdatacatalog** is available
- Expand the **dev** database
- Expand the **public** schema
- Review the Tickit sample tables including category, date, event, listing, sales, users, and venue



- Navigate to **Notebooks**
- Open the Tickit sample notebook
- Create a duplicate copy of the notebook by right-clicking it.
- Rename or save the notebook as **tickit-sample-notebook-for-cluster**



- Run the sample SQL queries from the notebook
- Observe the error indicating that the Tickit schema does not exist

```
SELECT eventname, total_price
FROM (SELECT eventid, total_price, ntile(1000) over(order by total_price desc) as percentile
```

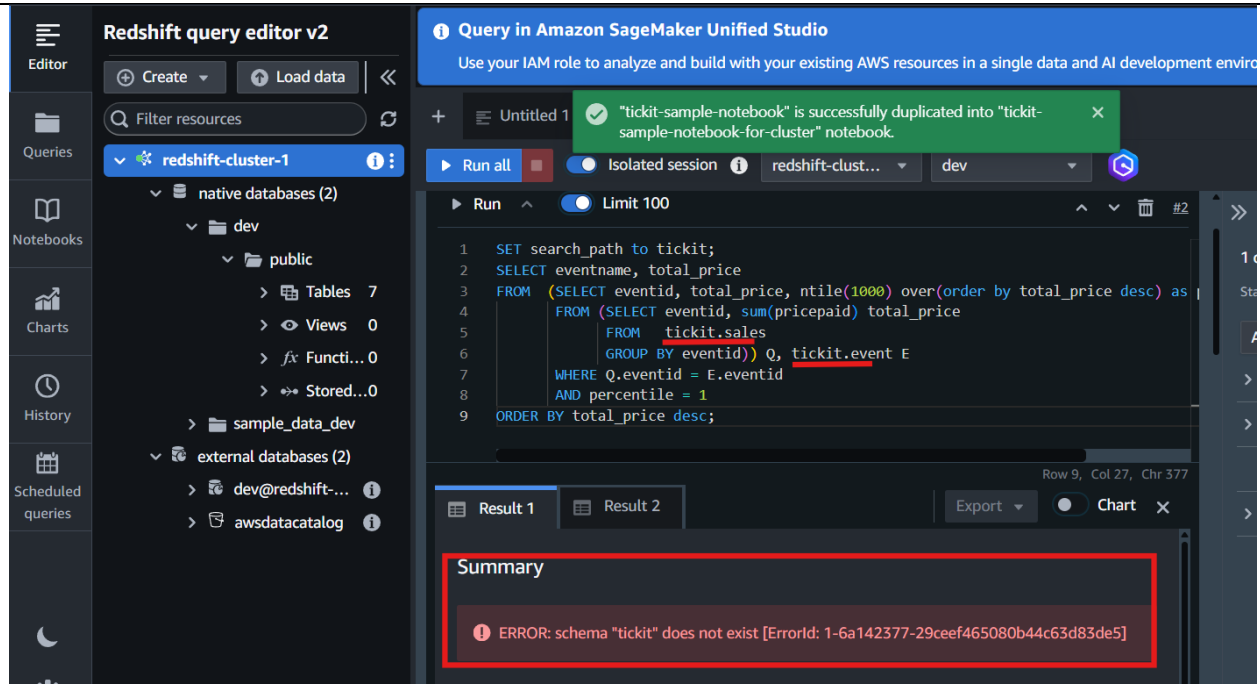
```

FROM (SELECT eventid, sum(pricepaid) total_price
      FROM tickit.sales
      GROUP BY eventid)) Q, tickit.event E
WHERE Q. eventid = E.eventid
AND percentile = 1
ORDER BY total_price desc;

```

- Review the error message generated by the query

schema "tickit" does not exist



- Identify that the Tickit tables are stored in the **public** schema instead of the Tickit schema
- Remove the search_path statement from the queries
- Remove the tickit schema prefix from table references
- Execute the corrected query using the public schema tables

```

SELECT eventname, total_price
      FROM (SELECT eventid, total_price, ntile(1000) over(order by total_price desc) as
            percentile
            FROM (SELECT eventid, sum(pricepaid) total_price
                  FROM sales
                  GROUP BY eventid)) Q, event E
WHERE Q. eventid = E.eventid
AND percentile = 1
ORDER BY total_price desc;

```

Redshift query editor v2

Query in Amazon SageMaker Unified Studio

Use your IAM role to analyze and build with your existing AWS resources in a single data and AI development environment

Filter resources

redshift-cluster-1

native databases (2)

dev

public

Tables 7

Views 0

Functions 0

Stored procedures 0

sample_data_dev

external databases (2)

dev@redshift-...

awsdatacatalog

Query editor

Run all

Isolated session

redshift-clust...

dev

Run

Limit 100

```

1 SELECT eventname, total_price
2 FROM (SELECT eventid, total_price, ntile(1000) over(order by total_price desc) as
3       FROM (SELECT eventid, sum(pricepaid) total_price
4             FROM sales
5             GROUP BY eventid)) Q, event E
6 WHERE Q.eventid = E.eventid
7       AND percentile = 1
8 ORDER BY total_price desc;

```

Result 1 (9)

eventname	total_price
Adriana Lecouvreur	51846
Janet Jackson	51049
Phantom of the Opera	50301
The Little Mermaid	49956
Citizen Cope	49823
Sevendust	48020
Electra	47992

- Run the sales-per-event validation query
- Verify that the query returns event names and total sales information
- Run the quantity-per-buyer validation query
- Verify that the query returns ticket purchase quantities for buyers
- Confirm that the queries execute successfully without schema errors

Redshift query editor v2

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Query editor

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Isolated session

redshift-clust...

dev

Run

Limit 100

```

1 SELECT firstname, lastname, total_quantity
2 FROM (SELECT buyerid, sum(qtysold) total_quantity
3       FROM sales
4       GROUP BY buyerid
5       ORDER BY total_quantity desc limit 10) Q, users
6 WHERE Q.buyerid = user_id
7 ORDER BY Q.total_quantity desc;

```

Result 1 (10)

firstname	lastname	total_quantity
Jerry	Nichols	67
Kameko	Bowman	64
Armando	Lopez	64
Kellie	Savage	63
Herrod	Sparks	60
Rhona	Sweet	60